

ME350 FA 2025 Section 002

Lab #11

Monday 12:30 PM - 2:30 PM

Haoran Shi

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Office Hours: 3:00 PM - 5:00 PM Monday (Zoom)



Reminders

- Homework 6 is due **Thursday**, November 6th
- **Final Testing is due on December 3rd at midnight**
- **Do not prop the Mechatronics lab door open if there is no GSI/IA present**
 - Student access to the lab will be revoked if this happens



Extra In-Person Office Hours

- GSIs/IAs/Mike will each be hosting a two hour Mechatronics office hours session during the weeks of November 10th and November 17th
- **Each team can sign up for ONE session [here](#)**
- You do not need to pick your GSI/IA's session

Week of November 10th:

- Tuesday 11/11 9:30 am - 11:30 pm Nana Porter-Honick
- Tuesday 11/11 3:30 pm - 5:30 pm Haoran Shi
- Thursday 11/13 11:30 am - 1:30 pm Chloe Affaki
- Friday 11/14 10:00 am - 12:00 pm Lawrence Chen

Week of November 17th:

- Tuesday 11/18 9:00 am - 11:00 am James Lorenz
- Wednesday 11/19 10:30 am - 12:30 pm Moeen Radgolchin
- Tuesday 11/25 1:30 - 3:30 pm Mike

***check the office hour calendar as these will replace a GSI/IAs typical OH for that week**



	Nov 9	Nov 10	Nov 11	Nov 12	Nov 13	Nov 14	Nov 15
Lecture			Lecture 21 - Fasteners (MU)		Lecture 22 - Bearings Part 1 (MU)		
Homework					- Homework 7 (Controls) Due		
Lab		Lab 11 - Sensors and Microcontrollers 4: State Machines Lab Assignment #10 Due			Lab 12 - Project Work	Lab 12 - Project Work	
Project Timeline							
	Nov 16	Nov 17	Nov 18	Nov 19	Nov 20	Nov 21	Nov 22
Lecture			Lecture 23 - Bearings Part 2 (MU)		No Lecture		
Homework			Homework 8 Released				
Lab		Lab 12 - Project Work			Lab 13 - PRELIMINARY TESTING of PROJECT	Lab 13 - PRELIMINARY TESTING of PROJECT	
Project Timeline							
	Nov 23	Nov 24	Nov 25	Nov 26	Nov 27	Nov 28	Nov 29
Lecture			No Lecture	Thanksgiving Break			
Homework							
Lab	Lab 13 - PRELIMINARY TESTING of PROJECT		(Thanksgiving Recess Begins at 5pm)				
Project Timeline							



Agenda

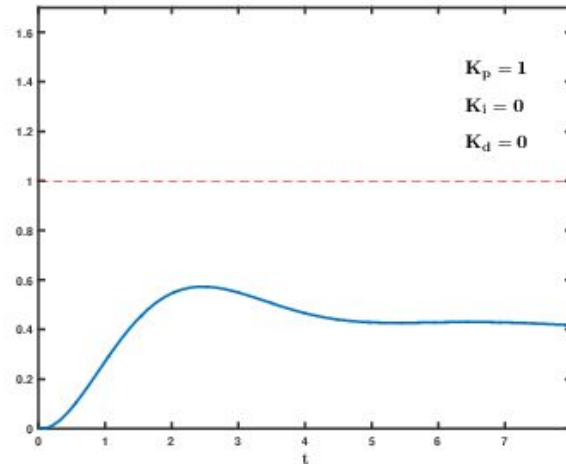
- PID Control Summary
- State Machine Diagram
- Full Linkage code tips
- Project Specifications
- Final Testing
- Final Assignments



PID Summary

Effects of *increasing* a parameter independently

Parameter	Rise time	Overshoot	Settling time	Steady-state error
K_p	Decrease	Increase	Small change	Decrease
K_i	Decrease	Increase	Increase	Eliminate
K_d	Minor change	Decrease	Decrease	No effect in theory



PID Summary

Caveat: Kd does not always kills oscillations

In this project, you have a low Kd value when

- The linkage is oscillating back and forth about the target lane position at a “low frequency”

But

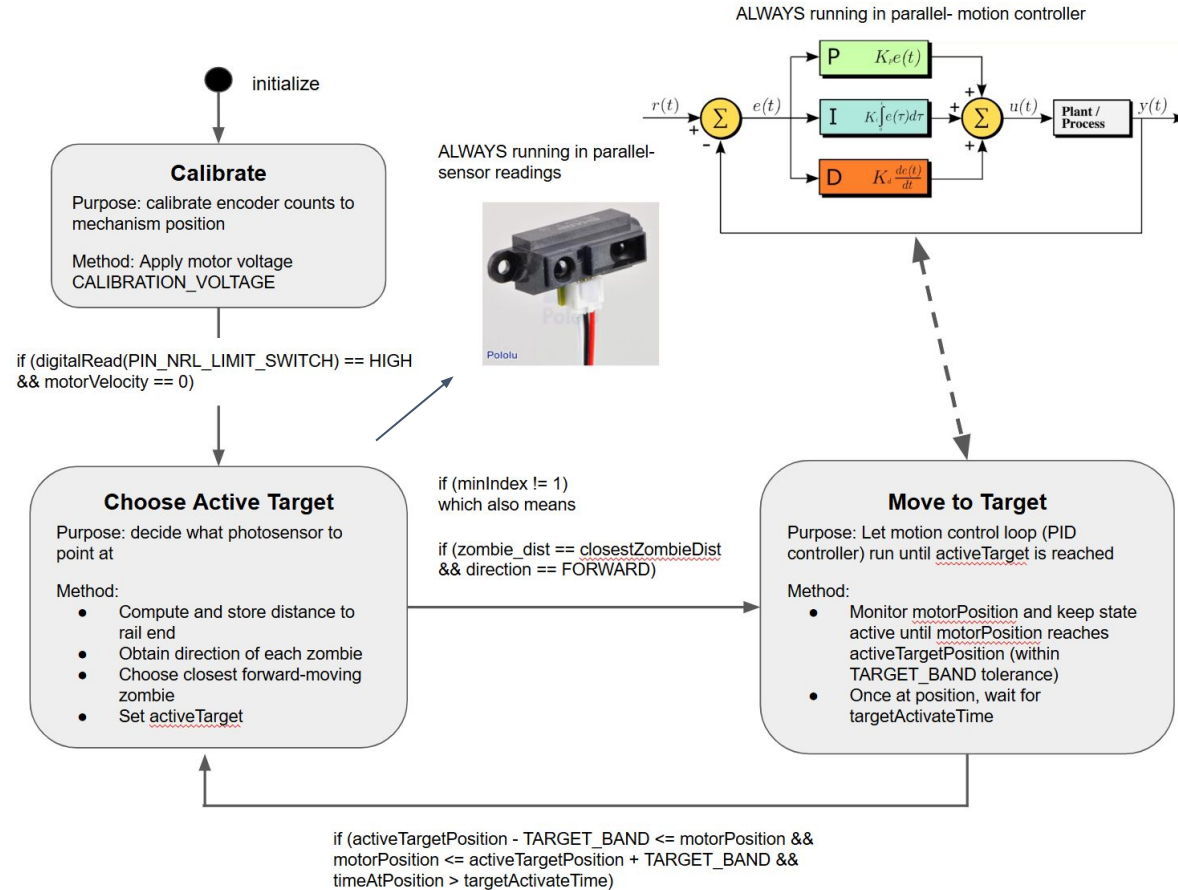
In this project, you have a high Kd value when

- If you observe “high frequency” juddering which is not necessarily about the target position but can be somewhere before that
- It will look like something is resisting the motion, the linkage is struggling to reach the target position while doing small but high frequency oscillations

For more information read [**PID Controllers Handout**](#)



State Machine Diagram



Full Linkage Code Tips

- Read through the whole code and understand it first before you try running the linkage!
 - See problem #1 (Arduino Scavenger Hunt) in HW7
- **Speed is everything!**
 - Comment out the serial print statements once you have a working linkage code and feel comfortable tuning values without it to get higher refresh rates on the speed controller
 - Try reducing the delays/wait times to increase how fast you move
 - If you still aren't happy with the speed, try changing the structure of the code
- If you have any questions about the code or want to discuss ideas, feel free to ask



Project Specifications

Table 1: Evaluation criteria.

Specification	Description	Target	% of Grade	Method of Measurement
Transmission Angle Deviation	$ Transmission\ Angle - 90^\circ $ (for the largest angle)	0	20	Machinist's Protractor
Weight		Minimize	30	Scale
Performance Score	Based on number of targets dropped and amount of light on targets	Maximize	50	Playing Field

Table 2: Performance Scoring.

Action	Measurement	Scoring
Shine Light	Sum of analogue reading of each photosensor in each stage	Measurement / 1000
Drop Target	Number of targets dropped	Measurement $\times$ 10

Project Description: [\[CLICK HERE\]](#)



Final Testing

- Submit a video of your team's best run to get a performance score. The video must have:
 - Calibration - your linkage should **start from a random position** set by hand in the video, and then it should automatically move to the limit switch to calibrate
 - Round 1, round 2 and round 3 in one continuous video
 - Final score displayed on the playing field

Missing any of the above can result in a 0 performance score!



Final Testing

- Move your mechanism to an arbitrary position and turn on your arduino to calibrate the linkage.
- To begin the round, press the start button on the playing field
- The flags on the playing field will start approaching the target once each round is started. As your mechanism shines light on a target, you should see the corresponding flag returning to its original position.
- The testing consists of three rounds in total. In the first two rounds, the flags continue approaches the target until reaching time limit. The last round stops once a flag reaches the end of its row.
- To restart the playing field, unplug it and plug it back in.



Final Testing

- At the end of your run, your score will be displayed



Lab 12 Assignment: Return Project Kits

- Return your project kit to your cabinet/shelf by December 4th at Midnight
- Your box should contain:
 - Your fully assembled mechanism
 - Your fully wired Arduino
 - All purchased BOM items
 - All provided kit materials
- Your box should **NOT** contain:
 - Tools from the X50 Assembly Room or Mechatronics Room
 - Extra stashed parts like shoulder bolts, playing field bolts, washers

***Returning a disassembled mechanism, unwired Arduino/H-Bridge/Breadboard, or turning in your kit after December 4th will result in a 0% Craftsmanship grade**

****Containing tools or extra stashed parts in your box will have an adverse effect on your Craftsmanship grade**



Prototype Craftsmanship (20% of final project grade)

- I will be grading the overall design, quality of workmanship, surface finish, tolerance and fit, and aesthetics of your final design
- Below are some (but not all) things I will check to determine your grade:
 - Do parts have sharp edges or burrs?
 - Are there more than two unnecessary features (i.e. extra holes)?
 - Is everything machined correctly?
 - Is there excessive play or friction at the joints? Are shoulder bolts loose?
 - Is the circuitry professional (i.e. wire braiding, shrink tube, soldering are appropriately used and breadboard connections are solid)?
 - Is the limit switch rigidly attached?
 - Is the flashlight holder mounting solid?



Today

- State Machine [Lab 11 Tutorial](#)
 - ***Keep track of your design parameters as you tune.***

Due by end of the semester

1. [Sign up](#) for extra lab hours in the Mechatronics lab
2. Submit final testing video (Due Dec 3rd by midnight)
3. Lab Assignment 12: Return project kit to X50 Assembly Room (Due Dec. 4th)
4. Final Design Report (Due Dec. 8th at midnight) - [template](#)